GRANT FELLOWS

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EDUCATION

Bachelor of Science in Mechanical Engineering, Purdue University — Expected Dec 2027

SKILLS

3D Modeling & Simulation	SolidWorks, Siemens NX, Inventor, Fusion 360, Simcenter/FEMAP (CAD/CAM/FEA); GD&T (ASME Y14.5)
Fabrication	Manual mill/lathe; CNC mill (Tormach), 3D printing, laser cutting
Software Development	Python, Java, JavaScript, MATLAB; Flask, HTML, CSS; ROS 2

EXPERIENCE

Mechanical Engineering Intern, Anyware Robotics — Aug 2026 – Present
Fremont, California

- CAD design, R&D, analysis, prototyping, fabrication, and testing of vacuum gripper innovations to improve end-effector performance and payload capacity
- Design analysis and optimization with Python scripts; drawings to ASME Y14.5 GD&T

Undergraduate Research Assistant, Applied Optics Lab, Purdue University — May 2026 – Aug 2026
West Lafayette, Indiana

- Designed, fabricated, and tested a working prototype 3-axis stepper-driven gantry robot with servo gripper for automated biological sample handling in Prof. Euiwon Bae's group

Software and Autonomy Lead, Purdue Vertical Flight Systems — Aug 2025 – May 2026
West Lafayette, Indiana

- Led the autonomy sub-team developing ROS 2 nodes for flight control, sensor integration, and obstacle avoidance on an autonomous emergency response aircraft (GoAERO Prize); built obstacle detection pipeline from stereo camera point clouds

OrbShip Fellow, Orb Aerospace — May 2025 – Aug 2025
Grand Rapids, Michigan

- Developed a Python library and web app improving FEM simulation of additively manufactured parts, letting Orb engineers predict stiffness and failure of printed parts before physical testing; presented approach to Orb team
- Conducted 50+ ASTM tensile tests and data analysis to demonstrate validity of simulation in real-world parts

Systems Engineering Lead, Purdue Lunabotics — Aug 2024 – May 2025
West Lafayette, Indiana

- Led systems engineering for Purdue's entry into the NASA Lunabotics competition, tracking and verifying system requirements; placed 5th in Systems Engineering, and 3rd at the UCF Qualifier (2025)

PUBLICATIONS

X. You, V. Cattaneo, S. Yanagisawa, S. B. Larson, M. C. Lee, R. R. Kurup, **G. Fellows**, A. K. Swadi, A. R. Chandra, and T. A. Guimarães, "Development of a High-Payload Electric Vertical Takeoff and Landing Multirotor for the GoAERO Competition," *AIAA AVIATION Forum*, 2026, doi:10.2514/6.2026-4585

PROJECTS

- A GPS mount for ship railings. Built for a container ship captain's custom use case; parts are in active use today
- An all-terrain rover inspired by NASA Mars rover suspension designs (github.com/gwfellows/rover)